DIGITAL FORENSICS – I

Unit 2: Digital Evidence & Acquisition
Topics 2.2 & 2.3 — Types and Characteristics of Digital Evidence, Chain of Custody

2.2 Types and Characteristics of Digital Evidence

Digital evidence does not come in a single uniform format — it varies widely depending on the device, the state of the system at the time of collection, and how long the data persists after the event that created it. Understanding these types and characteristics allows an investigator to prioritize what to collect first and choose the right acquisition method for each.

1. Types of Digital Evidence (by Volatility)

The most fundamental classification of digital evidence is based on how easily it is lost once a system is powered off, disconnected, or left running.

(a) Volatile Evidence

Data that exists only while the system is powered on and is lost the moment power is removed. It must be captured live, before shutdown.

- RAM contents (running processes, open files, decryption keys in memory)
- Active network connections and open ports
- Logged-in user sessions and clipboard contents
- Contents of CPU cache and registers

(b) Non-Volatile Evidence

Data that persists even after the system is powered off, since it is stored on physical media.

- Hard disk / SSD contents (files, folders, deleted data remnants)
- System logs, event logs, application logs
- Registry entries, configuration files
- Backup files, archived emails, cloud-synced data

2. Types of Digital Evidence (by Source)

Source Type	Examples of Evidence
End-user devices	Documents, photos, browser history, chat logs, deleted files
Network evidence	Packet captures, firewall logs, IDS/IPS alerts, DNS logs
Server/application evidence	Web server logs, database records, authentication logs
Mobile evidence	Call logs, SMS, app data, GPS location history
Cloud evidence	Account activity logs, stored files, synced data, VM snapshots
IoT/embedded evidence	Smart device logs, sensor data, wearable device history

3. Order of Volatility (RFC 3227)

RFC 3227 provides a standard sequence for collecting evidence, starting with the most volatile data first so nothing critical is lost during acquisition:

- 1. CPU registers, cache
- 2. Routing table, ARP cache, process table, kernel statistics, RAM
- 3. Temporary file systems
- 4. Disk (hard drive / SSD)
- 5. Remote logging and monitoring data related to the system
- 6. Physical configuration and network topology
- 7. Archival media (backups, offline storage)

4. Characteristics of Digital Evidence

- **Fragility:** Can be altered, corrupted, or destroyed easily by improper handling, making careful acquisition essential.
- **Latency:** Some evidence (like RAM) is time-sensitive and disappears within seconds to minutes if not captured.
- **Duplicability:** Unlike physical evidence, digital evidence can be copied bit-for-bit, producing an identical working copy for analysis.
- **Volume:** Digital evidence can exist in massive quantities (terabytes of logs/data), requiring efficient filtering and search techniques.
- **Cross-jurisdictional nature:** Evidence, especially cloud-based, may be stored across multiple countries, raising jurisdiction and legal-access challenges.
- **Invisibility to the naked eye:** Requires specialized tools/software to view, interpret, and validate the evidence.
- **Metadata dependency:** Much of the evidentiary value lies in metadata (timestamps, author info, file properties), not just visible content.

Exam Tip: A frequent 4/5-mark question asks students to "explain the characteristics of digital evidence." Use the acronym-friendly list above (Fragility, Latency, Duplicability, Volume, Jurisdiction, Invisibility, Metadata) — it's easy to recall and covers the expected breadth.

2.3 Chain of Custody

1. Definition

Chain of Custody (CoC) is the chronological, documented record that traces the seizure, custody, control, transfer, analysis, and disposition of evidence — physical or digital. It proves that the evidence presented in court is the same evidence that was originally collected, and that it has not been altered, tampered with, or substituted at any point.

2. Why Chain of Custody Matters

- Establishes the authenticity and integrity of evidence for legal admissibility.
- Protects against claims of evidence tampering, mishandling, or contamination.
- Provides accountability — every person who touched the evidence is identifiable.
- A broken or incomplete chain can lead to evidence being ruled inadmissible in court, regardless of its actual content.

3. Key Elements Recorded in a Chain of Custody Form

Field	Purpose
Case/Reference Number	Uniquely identifies the investigation the evidence belongs to
Description of Evidence	Device type, serial number, make/model, unique identifiers
Date & Time of Collection	Establishes exactly when evidence was seized
Location of Collection	Documents where the evidence was physically found
Collected By (Name/ID)	Identifies the person who first acquired the evidence
Hash Value (MD5/SHA-256)	Cryptographic fingerprint to prove integrity at each stage
Transfer Log	Records every handover: from whom, to whom, date, time, reason
Storage Location	Where and how the evidence is securely stored between transfers
Signatures	Signed acknowledgment by each custodian at every transfer point

4. Role of Hashing in Chain of Custody

Hashing is the primary technical mechanism used to prove that evidence has not been altered. When evidence is first acquired, a cryptographic hash (commonly MD5 or SHA-256) is generated for the original media and again for the forensic image created from it. If both hash values match, it proves the copy is identical to the original. This same hash is recalculated at every subsequent stage of examination — if it ever changes, it indicates the evidence has been modified, intentionally or accidentally.

5. Steps in Maintaining Chain of Custody

- 1. Document the scene and evidence condition before collection (photographs, notes).
- 2. Seize evidence using proper protective measures (write-blockers, anti-static bags, Faraday bags for mobiles).
- 3. Generate a hash value immediately upon acquisition.
- 4. Label and tag evidence with a unique identifier linked to the case file.

- 5. Log every transfer of custody with date, time, names, and reason for transfer.
- 6. Store evidence in a secure, access-controlled location when not actively being examined.
- 7. Re-verify the hash value before and after analysis to confirm no changes occurred.
- 8. Maintain the completed chain of custody form as part of the final case file/report.

6. Consequences of a Broken Chain of Custody

- Evidence may be challenged and ruled inadmissible in court.
- Credibility of the entire investigation and the investigator may be questioned.
- Can result in case dismissal even when the underlying evidence is factually accurate.
- May expose the organization or investigator to legal liability for mishandling evidence.

Exam Tip: Chain of custody questions often ask students to "list the contents of a chain of custody form" or "explain the importance of chain of custody." Structure answers as: Definition → Importance → Key elements/fields → Consequences of failure. This progression covers both conceptual and applied aspects expected in 4/5-mark answers.